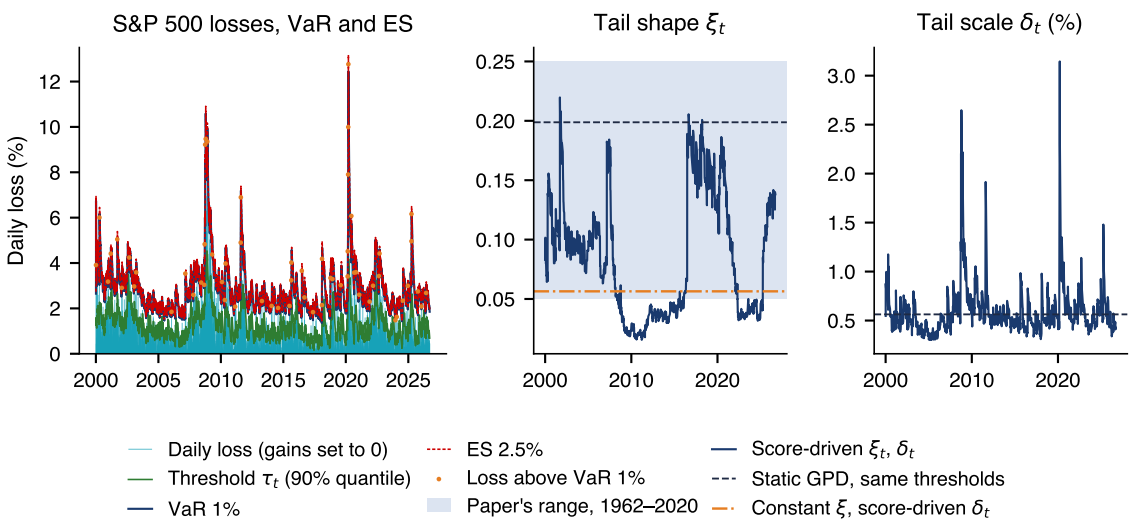

S&P 500 losses, VaR and ES
Daily loss (%)
0
2
4
6
8
10
12
2000
2005
2010
2015
2020
2025
Tail shape $\xi_t$
0.05
0.10
0.15
0.20
0.25
2000
2010
2020
Tail scale $\delta_t$ (%)
0.5
1.0
1.5
2.0
2.5
3.0
2000
2010
2020
Daily loss (gains set to 0)
Threshold $\tau_t$ (90% quantile)
VaR 1%
ES 2.5%
Loss above VaR 1%
Paper's range, 1962–2020
Score-driven $\xi_t$, $\delta_t$
Static GPD, same thresholds
Constant $\xi$, score-driven $\delta_t$